PROGRAM

```
import random
N = 8
# Calculate number of attacking pairs
def heuristic(state):
    attacks = 0
    for i in range(N):
        for j in range(i + 1, N):
            # Same row
            if state[i] == state[j]:
                attacks += 1
            # Same diagonal
            elif abs(state[i] - state[j]) == abs(i - j):
                attacks += 1
    return attacks
# Generate neighboring states
def get_neighbors(state):
    neighbors = []
    for col in range(N):
        for row in range(N):
            if row != state[col]:
                new_state = state.copy()
                new_state[col] = row
                neighbors.append(new_state)
    return neighbors
# Hill Climbing
def hill_climbing(start):
    current = start.copy()
    iterations = 0
    while True:
        current_h = heuristic(current)
        neighbors = get_neighbors(current)
        best = min(neighbors, key=heuristic)
        best_h = heuristic(best)
        if best_h >= current_h:
            break
        current = best
        iterations += 1
    return current, iterations
# Random Restart Hill Climbing
def random_restart():
    restart = 0
```

```
    while True:
    restart += 1
        start = [random.randint(0, 7) for _ in range(N)]
        solution, iterations = hill_climbing(start)
        if heuristic(solution) == 0:
            return solution, iterations, restart
# Main
random.seed(42)
start = [random.randint(0, 7) for _ in range(N)]
solution, iterations = hill_climbing(start)
print("Initial State :", start)
print("Hill Climbing Solution :", solution)
print("Attacking Pairs :", heuristic(solution))
print("Iterations :", iterations)
solution, iterations, restarts = random_restart()
print("\nRandom Restart Hill Climbing")
print("Solution :", solution)
print("Attacking Pairs :", heuristic(solution))
print("Iterations :", iterations)
print("Number of Restarts :", restarts)
```

OUTPUT

```
Initial State : [1, 0, 4, 3, 7, 5, 6, 2]
Hill Climbing Solution : [1, 0, 4, 3, 7, 5, 6, 2]
Attacking Pairs : 2
Iterations : 3

Random Restart Hill Climbing
Solution : [1, 7, 5, 0, 2, 4, 6, 3]
Attacking Pairs : 0
Iterations : 3
Number of Restarts : 13
```